

**Government of India
Ministry of Communication
Department of Telecommunications
Telecommunication Engineering Centre
Khurshid Lal Bhawan, Janpath, New Delhi – 110 001**

File No. 28-1/2022-MT/TEC

Dated: 20-09-2025

Subject: Minutes of the 5th meeting of NWG-13 corresponding to ITU-T Study Group-13 held on 19-09-2025.

The 5th Meeting of the NWG-13 corresponding to ITU-T Study Group 13- "Future Networks and Emerging Network Technologies" was held on 19-09-2025. List of participants is annexed (Annexure-I).

2. At the outset, Sh. Amit Kumar Srivastava, DDG (MT), TEC and Chairman of NWG-13, welcomed all the participants of NWG-13 and highlighted the goal of NWG-13. While briefing about the agenda of the meeting, he emphasised the timely submission of contributions by participants to NWG-13, preferably by 28.09.2025, so that it can be finalised in NWG meeting scheduled on 30.09.2025 and ensured necessary formalities required for submission of the contributions from Member State India in time to ITU-T SG-13 meeting scheduled from 28th October. He encouraged all members to continue their active participation and collaborative efforts in developing new contributions, aiming to achieve greater milestones in the upcoming ITU-T SG-13 meeting. He motivated members to come up with their ideas/work so that one-to-one meetings may also be conducted if so required for developing in proper contribution..
3. Thereafter, Sh. Sujit Kumar, Director (MT), TEC, started the discussion on the contributions received for consideration in NWG-13. He apprised all the members that four contributions (two on new work item proposals and two revisions of existing work items) had been received for discussion in the NWG-13 meeting.
4. The first contribution was presented by Ms. Shweta Kiran, IIT Bombay. She presented the revision of the existing work item on Draft Recommendation ITU-T Y.AI_HNO_ACPS — “Requirements and Framework for AI/ML-based Heterogeneous Network Optimization and Automated Communication Primitive Selection in Future Networks including IMT-2020 and beyond”. One of the members emphasised the need to cross-check the diagram in Section 8 of the contribution for corrections. The same was acknowledged for needful.
5. The second contribution was presented by Mr. Abhijan Bhattacharya, TCS. He presented the revision of the existing work item on Draft new Technical Report ITU-T YSTR.SemComGenAI — “Framework for Semantic Communication Pipeline with

- Generative AI-based Reconstruction”. One of the members raised a query regarding the reference mentioned in the added text. The same was acknowledged for needful.
6. The third contribution was presented by Mr. Abhijan Bhattacharya, TCS. He presented the new work item proposal on Y.RecMscNPN — “Recommendation for Architecture and Requirements for Mine Supply Chain (MSC) using a Combination of Public and Non-Public Networks”. One of the members raised a query regarding the addition of a reference architecture for the mine supply chain, which was clarified by Mr. Abhijan. It was also suggested by members to maintain a standard font throughout the document, as specified and seen in ITU-T documents . Another member suggested modifying one of the diagrams in the contribution. The same was acknowledged for needful..
 7. The fourth contribution was presented by Sh. Sujit Kumar, Director (MT-I), TEC. He presented the new work item on ITU-T Y.CNAP_RA — “Requirements and Architectural Framework for Calling Name Presentation (CNAP) Implementation in Developing Countries”. One of the members suggested changing the scope from “Requirement of CNAP” to “Requirement of CNAP in Developing Countries”, keeping in mind the gap analysis in the document. The same was acknowledged for needful.
 8. DDG (MT), TEC appreciated all the members for critically examining the documents submitted in such a short duration. He also acknowledged that for subsequent meetings, contributions should be shared with all members at least one day in advance so that they may review the documents and submit their suggestions and can be discussed in the NWG meeting..
 9. The Chair and Vice-Chairs of NWG-13 also briefed the new members about SG-13, ITU-T, and its focus areas. They informed members about the process of submitting contributions to ITU-T. They motivated all members to bring forward more innovative ideas which can be shaped into contributions for further submission to ITU-T. Further, they also mentioned that all contributions (new and existing work items) discussed today will be circulated among all NWG-13 members for their inputs and suggestions.
 10. In the end, DDG (MT) thanked all the participants for attending the meeting.

(Sujit Kumar)
Director (MT), TEC

Information to the participants:-

1. Regarding the upcoming meeting of SG-13, it is informed that the next ITU-T SG-13 meeting is scheduled to take place from 28th October – 6th November 2025 at Tashkent, Uzbekistan. All the members can go through the different questions of SG-13 ([List of Questions and Rapporteurs](#)) which are also annexed in annexure II and identify the questions which are aligned to their area of interest. It is also advised to the members to share their willingness to specific questions of SG-13 and share their work in order to build suitable contribution. For that, one to one meeting may also be conducted by NWG-13 management team to apprise and develop the ideas/concept to the contribution format. The details of questions of SG-13 along with their term of reference (TOR) are enclosed at **Annexure-II**. To access the on-going work, on-going documents, members requires ties account, and therefore members may contact us through phone/WA/email etc. for arranging the documents as per requirement for better understanding and developing the contribution.
2. It is to inform all the members that participation of delegates from member state India requires ties account and a mandatory undertaking. The template of the undertaking is as per **Annexure-III**. For creation of ties account, "[ITU User Account & TIES Services - ITU](#)" may be visited and informed along with undertaking. Based on contributions developed by the members, this office will approach competent authority for approval of Ties account required for participating and presenting the document in the ITU meetings.

Annexure- I**List of participants**

1. Amit Kumar Srivastava, DDG(MT), TEC
2. Sujit kumar, Dir (MT-I), TEC
3. Dhanesh Goel, Dir(MT-III), TEC
4. Jyoti Roat, Dir (MT-II), TEC
5. Drupad Gupta, ADET (MT), TEC
6. Kuldeep, DoT
7. Sandeep Singh, NCA-T
8. Suryash Gautam, DoT Gujarat
9. Dr. Rashmi Kamran, IIT Bombay
10. Preksha Shah, IIT Bombay
11. Shwetha Kiran, IIT Bombay
12. Anirban Bhowal, NIT Rourkela
13. Abhijan Bhattacharyya, TCS
14. SUSHANT KUMAR JHA, RA, MT, TEC
15. Dr. Radhika, PES University
16. Ajay Singh, IIT Jammu
17. Ankush Chaudhary, IIT Jodhpur
18. Dr Amit Kumar Singh, IIT Patna
19. Bodhibrata Mukhopadhyay, IIT Roorkee
20. Ankit Jain, BP
21. Arun Kumar Singh, IIT Jodhpur
22. Shiv, ITS India
23. BHARAT JYOTI RANJAN SAHU, SIKSHA O ANUSANDHAN UNIVERSITY
24. Dr Ashish Raman, NIT Jalandar
25. SANJAY BHARDWAJ, TEC
26. Shraddha Nand, DCIPHERS IT SOLUTIONS
27. P.Kaythry, SSN College of Engineering

Annexure-II**SG-13: Questions under study and their corresponding Term of references (ToR) and work Items (WI)**

Sr. No.	Question Under Study	Links for ToR & WI
1	Q1/13 (WP3/13) Future Networks: Innovative service scenarios, including environmental and socio economical aspects	Terms of reference Work Items
2	Q2/13 (WP3/13) Next-generation network (NGN) evolution with innovative technologies including software-defined networking (SDN) and network function virtualization (NFV)	Terms of reference Work Items
3	Q5/13 (WP3/13) Applying Future Networks and innovation in developing countries	Terms of reference Work Items
4	Q6/13 (WP1/13) Networks beyond IMT2020: Quality of service (QoS) mechanisms	Terms of reference Work Items
5	Q7/13 (WP2/13) Future Networks: Deep packet inspection and network intelligence	Terms of reference Work Items
6	Q16/13 (WP3/13) Future Networks: Trustworthy and Quantum Enhanced Networking and Services	Terms of reference Work Items
7	Q17/13 (WP2/13) Future Networks: Requirements and capabilities for computing including cloud computing and data handling	Terms of reference Work Items
8	Q18/13 (WP2/13) Future Networks: Functional architecture for computing including cloud computing and data handling	Terms of reference Work Items
9	Q19/13 (WP2/13) Future Networks: End-to-end management, governance, and security for computing including cloud computing and data handling	Terms of reference Work Items
10	Q20/13 (WP1/13) Networks beyond IMT-2020 and machine learning: Requirements and architecture	Terms of reference Work Items
11	Q21/13 (WP1/13) Networks beyond IMT-2020: Network softwarization	Terms of reference Work Items
12	Q22/13 (WP1/13) Networks beyond IMT-2020: Emerging network technologies	Terms of reference Work Items
13	Q23/13 (WP1/13) Networks beyond IMT-2020: Fixed, mobile and satellite convergence	Terms of reference Work Items

Annexure- III**UNDERTAKING**

I _____ hereby agree to abide by the guidelines issued by the DoT vide letter no. 7-10/2014-IR and also the following conditions for getting registered as a delegate as part of the DoT delegation to participate in the conference _____ (details)

1. The total expenditure of the visit will be borne by me or my organization and DoT does not have any financial or legal liability for the visit.
2. I will only express the official view as authorized by the Head of the Delegation or as approved by the Indian administration.
3. Once registered for a particular program/ event/ meeting/ conference as a part of official Indian delegation, I will not be a part of any other delegation for that particular program/ event/ meeting/ conference.
4. I will maintain the confidentiality, integrity and sanctity of the internal deliberations, discussions and strategies of the official Indian delegation.

Signature of the Member

(Details: Name, Position and Organisation)

To,

The Dy. Director General (IR)
Department of Telecommunication
Room No. 403, Sanchar Bhawan
New Delhi.